

ILAN JORDAN

Mechanical Engineering & Materials Science · Physics — Duke University
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PROFILE

I am drawn to physics for what it makes possible. The distance between a principle understood in a lab and a technology that changes how people live is where the interesting engineering happens, and it is where I want to spend my career. I study Mechanical Engineering & Materials Science alongside Physics because the problems I care about do not respect the boundary between them: understanding why a system behaves as it does and building something that survives contact with the real world are the same task approached from two directions.

Energy is what pulls at me most. I grew up in Brazil, where blackouts are ordinary enough that you plan around them, so reliable power has never been an abstraction to me. Fusion is the version of that problem I find hardest to look away from — plasma physics, materials under conditions nothing was designed for, control problems that are still open. I am also interested in machine learning applied to physical systems, using learned models for control and prediction where first-principles simulation is too slow or too incomplete to stand on its own.

EDUCATION

B.S.E. Mechanical Engineering & Materials Science; second major in Physics · Duke University · Durham, USA Aug 2025
– May 2029 (expected)

GPA 3.86 / 4.00

Relevant Coursework: Mechanics of Solids · Multivariable Calculus · Calculus II · Linear Algebra (Matrices & Vectors) · Electricity, Magnetism & Vector Calculus · Introductory Mechanics · Experimental Physics I · Probability · Core Concepts in Chemistry · Data Structures & Algorithms · Engineering Design & Communication · Engineering Innovation

Visiting coursework, Mechanical Engineering · Universidade de São Paulo, Escola Politécnica · São Paulo, Brazil 6 months, 2025

GPA 8.7 / 10

Secondary education, English Honours Programme · Colégio Bandeirantes · São Paulo, Brazil 2020 – 2024

GPA 9.6 / 10 (top 2% of class). Selected in ninth grade for the English Honours Programme, which admits the top 20% of English students.

Standardised tests: SAT 1520 (Mathematics 800, English 720) · TOEFL 106 (Reading 30, Listening 29, Writing 24, Speaking 23)

RESEARCH EXPERIENCE

Research Assistant, Metallurgy · Universidade de São Paulo, Escola Politécnica · São Paulo, Brazil Jun – Aug 2026

- Studied a railway-grade steel alloy (OT-90), calculating in Thermo-Calc the heat treatment required to convert it into martensite for improved hardness and wear resistance.
- Executed the treatment on a Bähr dilatometer and verified the resulting OT-90 microstructure by X-ray diffraction, confirming the predicted transformation.

Research Assistant · Duke Physics Neutrino Lab · Durham, USA Mar 2026 – present

- Analyse photosensitive detector data for a range of cosmic particles alongside field specialists, supporting ongoing neutrino detection research.

PROFESSIONAL EXPERIENCE

Development Specialist I · Timpel Medical · São Paulo, Brazil May – Aug 2026

- Led development of EITSIM Studio, a full-featured Electrical Impedance Tomography simulator reproducing ICU respiratory mechanics and hemodynamics in real time, used for hardware-in-the-loop verification of EIT medical devices, clinical research, and academic teaching.
- Built the application end-to-end in Python (PySide6/Qt6, PyQtGraph) over a high-fidelity C++ physics engine, delivering a responsive ICU-monitor interface with live integration to a Hamilton C6 ventilator driving the simulated lung.
- Emulated the EIT acquisition module so that a real, unmodified clinical monitor renders live imagery driven entirely by the simulator, with no physical phantom required, using a forward model grounded in applied linear algebra.
- Built the clinical analysis suite — PEEP titration, pressure–volume and lung-recruitment curves — and authored the system’s technical documentation.

- Replaced live-animal (porcine) testing in device verification, saving an estimated US\$4,000 per preclinical validation campaign.

Web Developer & AI Optimisation Intern · TutorMundi · São Paulo, Brazil *Jul – Aug 2024*

- Debugged, developed features for, and quality-tested AI responses in CheckLLM, a tool that validates an education app's answers against a large language model.
- Worked in JavaScript, HTML, CSS, Meteor, and Git within a twenty-person web development team, on a framework reaching 50 schools nationally.

TEACHING

Grader, Linear Algebra · Duke University, Department of Mathematics · Durham, USA *Aug 2026 – present*

- Evaluate weekly problem sets and examinations, delivering written feedback on proof structure and computational accuracy and keeping grading consistent across sections.

Teaching Assistant, Calculus II · Duke University · Durham, USA *Jan 2026 – present*

- Run four weekly help-room hours for a course of over 300 students, covering integration techniques, sequences and series, and differential equations.

Head Physics Coordinator (2023–24); Subject Monitor (2021–22) · ARPA revision programme, Colégio Bandeirantes · São Paulo, Brazil *2021 – 2024*

- Selected by faculty to oversee the production of teacher-approved subject summaries each term, and taught online and in-person review sessions for over 250 students.
- Coordinated a team of student monitors across physics, chemistry, and mathematics, standardising review material across three grade levels.

ENGINEERING DESIGN & SIMULATION

CAD & Simulation, Liquid Propulsion Team · Duke AERO · Durham, USA *2025 – present*

- Build CAD models of engine and feed-system components and run Ansys fluid simulations of combustion and regenerative cooling for a student-built liquid rocket engine targeting 50,000 ft.
- Coordinate across nine engineering subteams — propulsion, avionics, structures — to keep interfaces consistent between designs and reconcile them ahead of test campaigns.

Chief Engineer, Automatic Paintbrush Cleaner (EGR101) · Duke University · Durham, USA *Aug - Dec 2025*

- Researched, conceived, designed and built a functioning prototype of a market-ready automatic paintbrush cleaner, owning the mechanical design through to final build.
- Directed design reviews and user testing with four teammates, validating cleaning effectiveness and ease of use against manual washing.

Combustion Engine Programme · Instituto Mauá de Tecnologia · São Paulo, Brazil *2023*

- Completed university-level coursework on engine cycles, stoichiometry, efficiency, rich and lean fuel mixtures, and engine mechanisms.
- Tuned fuel composition and modular gearing on a live test engine for the final project, maximising power output while preserving torque.

Robotics & 3D Printing, Summer Discovery · UCLA · Los Angeles, USA *2023*

- Completed twenty hours per week of coursework in robotics and additive manufacturing alongside UCLA faculty over three weeks.
- Modelled parts in Onshape and operated laser cutting and metalworking equipment to fabricate functioning assemblies.

INDEPENDENT PROJECTS

Radio-Controlled Aircraft · Designer and builder *2024*

- Designed, 3D-printed and flew a radio-controlled aircraft over ten weeks, handling airframe CAD, control surfaces, and electronics integration.

Homemade Electric Generator · Designer and builder *2022 – 2023*

- Built a working electric generator from scratch under the guidance of a local university professor, investing roughly ten hours per week across five weeks.

Automatic Irrigation System · Designer and builder *2024*

- Built an Arduino-driven irrigation controller with moisture sensing and scheduled watering logic to automate plant care.

Divisions and Dragons · Solo developer, JavaScript browser game 2022

- Developed a single-player RPG-style browser game from scratch in JavaScript, later published on the school website.

CAD and Fabrication Portfolio · Designer 2022 – present

- Modelled and fabricated an electric truck, a solar panel model, and a functioning Buzz Lightyear costume using Onshape, Siemens NX, and 3D printing.
- Produced a catalogue of printed design pieces to build fluency across modelling workflows.

LEADERSHIP & COMMUNITY SERVICE

Academic & Professional Affairs Analyst · BRASA — Brazilian Student Association · Durham, USA 2025 – present

- Organise academic and professional events with industry leaders and professors for a network of 70+ Brazilian students in the Duke community.
- Lead outreach bringing in speakers across sectors — including Bain & Company, SAP, Ambev, and Dr Miguel Nicolelis — to discuss research, innovation, and career development.

Python Teacher · Unidos de Paraisópolis · São Paulo, Brazil 2024 – 2025

- Adapted lesson plans and built presentations to teach basic logic, mathematics and Python to students aged 12 to 25 in a local favela.
- Guided students through three working builds — a calculator, an interactive rock-paper-scissors game, and a digital bank — to reinforce algebra, geometry, and programming logic.

Project Co-Founder · Educational Access Club · São Paulo, Brazil 2024

- Co-created a web-based directory covering the Brazilian 8th–11th grade core curriculum, making it accessible to an NGO serving over 4,000 under-resourced teenagers and adults.
- Supervised the mathematics, history, geography and philosophy sections and coordinated material development with a partner organisation.

Volunteer · Shalom Community Centre · São Paulo, Brazil 2022 – 2023

- Managed inventory and verified meal counts for city-wide deliveries, adjusting preparation quotas per cycle to ensure every recipient was served.

HONOURS & COMPETITIONS

- Brazilian Physics Olympiad (OBF), 2023–24 — Silver medal (2023) and Bronze medal (2024); top 1.5% of over 40,000 participants.
- Kangaroo Mathematics Olympiad, 2021–24 — Gold medal (2023) and Silver medals (2022, 2024); over 30,000 participants.
- Brazilian Astronomy & Astronautics Olympiad (OBA), 2021–22 — two Gold medals; over 100,000 participants.
- Brazilian Geography Olympiad (GeoBrasil), 2022 — Gold medal, top 1% of over 15,000 participants.
- Brazilian Robotics Olympiad (OBR), 2024 — Bronze medal; over 40,000 participants.
- Unicamp Mathematics Olympiad, 2024 — reached the second round, top 3% of 10,000 participants.
- Brazilian Computing Olympiad (OBI), 2021 — reached the second round, top 10% of 12,000 participants.
- São Paulo State Chemistry Olympiad (OQSP), 2022–24 — Honourable Mentions for original scientific papers.

Olympiad training teams · Colégio Bandeirantes 2021 – 2024

- Attended four hours of weekly advanced physics and mathematics problem sessions across 25 weeks per year to represent the school nationally, competing in more than five national and regional competitions and earning nine medals.

TECHNICAL SKILLS

CAD & design: Onshape, Siemens NX, technical drawing

Simulation & analysis: Ansys (Fluent), Thermo-Calc

Laboratory: Bähr dilatometer, X-ray diffraction, photosensor data analysis

Fabrication: 3D printing, laser cutting, basic metalworking, Arduino

Programming: Python — advanced (PySide6/Qt6, PyQtGraph); JavaScript, HTML, CSS, Java — intermediate; Git and GitHub, Meteor

Other software: Microsoft Office 365 including Excel

Certifications: Claude Code 101

OTHER ACTIVITIES

Debate Club · Colégio Bandeirantes 2022

- Analysed current events, logical fallacies and debate theory in a 30-student group under geopolitics faculty, competing in team debates across diverse topics.

JavaScript Learning Programme · Colégio Bandeirantes 2022

- Selected for a competitive school programming track covering coding, game development and programming logic across 35 weeks.

LANGUAGES

Portuguese (native) · English (C2, fluent) · Spanish (B2, intermediate–advanced) · German (beginner)

INTERESTS

Piano, ukulele and music theory · running, basketball and tennis · STEM building projects · philosophy and debate · film, rock and jazz · tabletop and console games.